

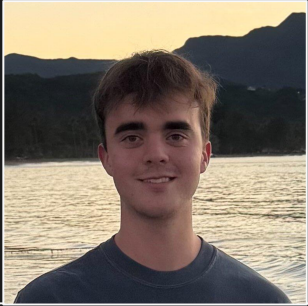
TRUE FORCE ARCADE RACK

Team:
sdmay26-40

Advisor:
Matt Post



Members



Jacob Garcia:
Software
Developer



Fadi Masannat:
Software
Developer



Sofi Gutierrez:
Software
Developer



Colin Yuska:
Electronics
Manager



Dylan Longlett:
Software
Developer &
Electronics



Andy Drafahl:
Software
Developer &
Team Manager



Parnika Dasgupta:
Software
Developer &
Client Liaison

Design Problem and Context

The Problem: Traditional strength testing machines are outdated

Context:

- Public venues
- Quick engagement
- Random users

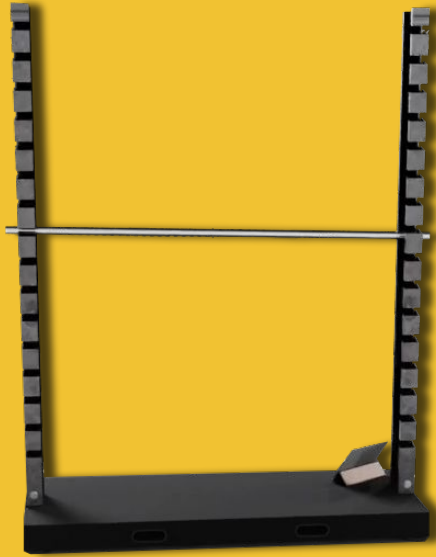


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What We Built

Before:



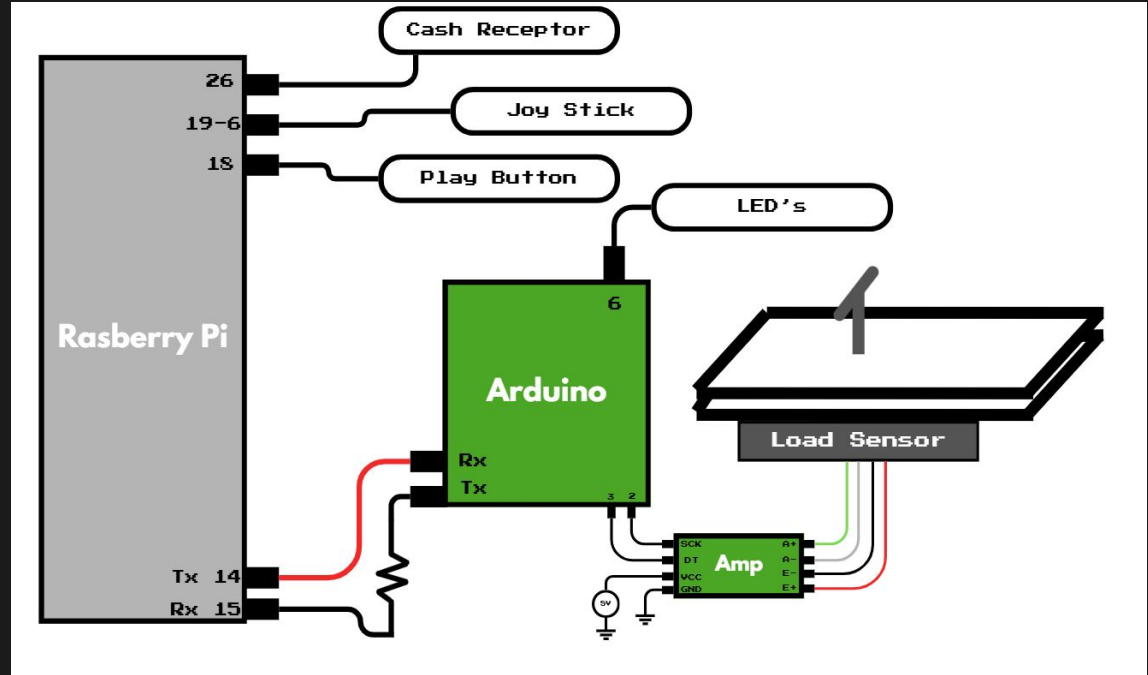
After:



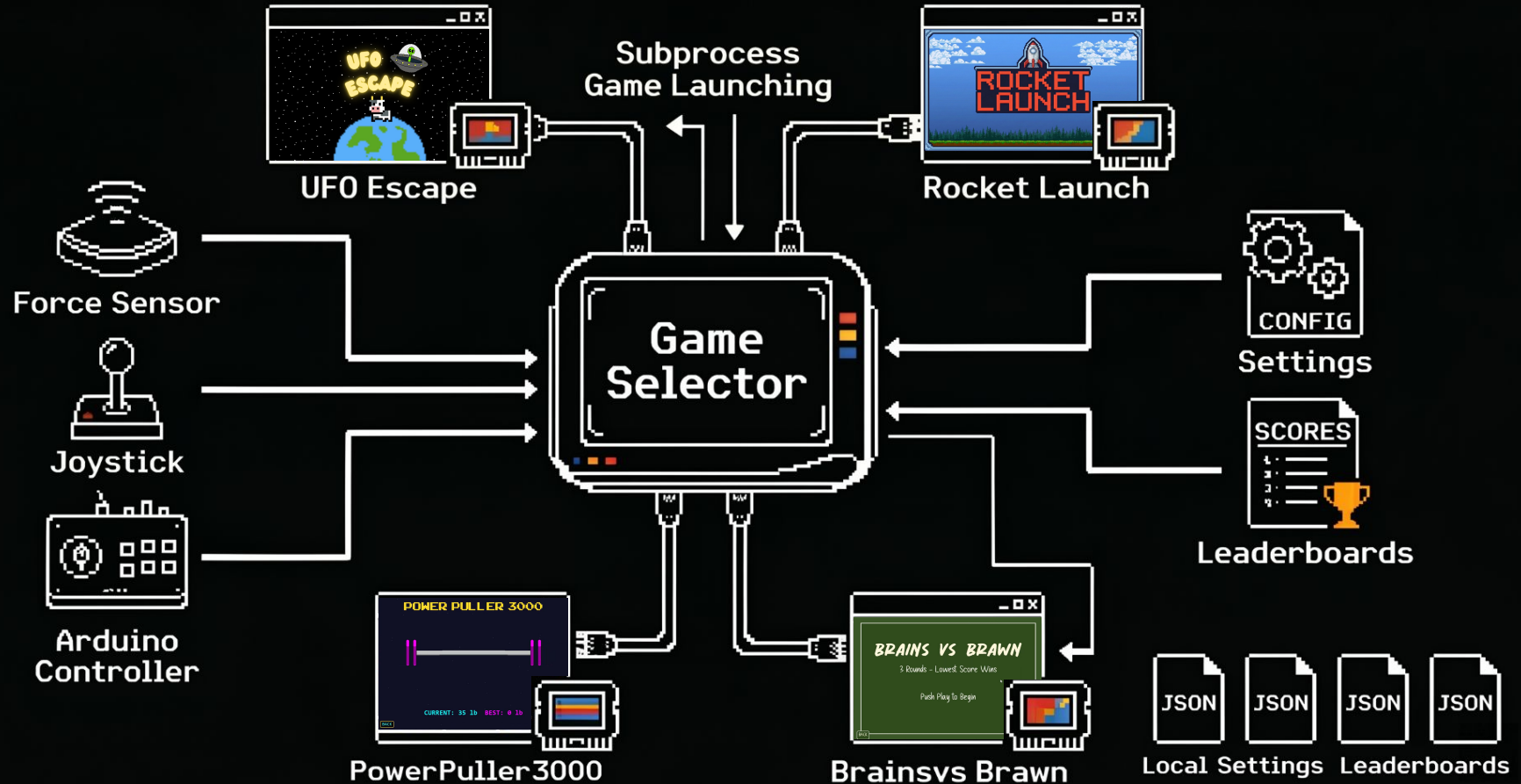
Hardware Overview

Features:

- Raspberry Pi $\leftrightarrow$ Arduino
- LEDs
- Cash Acceptor
- Load Sensor
- Cabinet Screen
- Play Button
- Joystick



TRUEFORCEARCADE MODULAR SOFTWARE ARCHITECTURE



TRUEFORCE ARCADE CODEBASE DESIGN

Pages / Screens



GameSelector



Settings Page



Results Screen



Leaderboards



Name Entry

Game Modules



Power Puller



Rocket Launch



UFO Escape



Brains vs Brawn

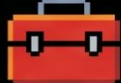


Balloon Burst

Shared Utilities



WeightConnection



ButtonControl



Settings Loader



JSON Persistence



Asset Loading

Dev Tools



Pytest Suite



Deploy Scripts

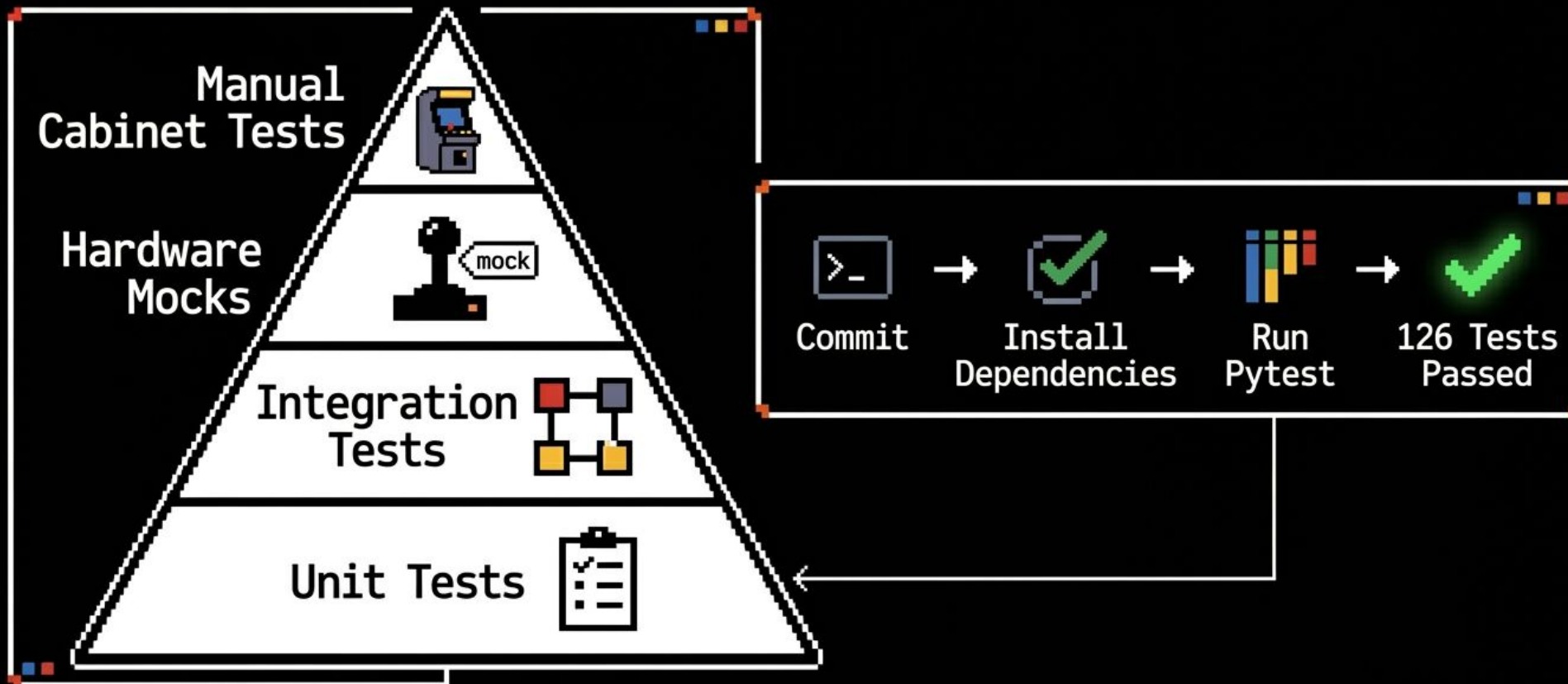


Asset Tools



Docs

TESTING + CI PIPELINE



Fast feedback before Pi deployment

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Authentic Retro Music and Sound Effects



Custom 16-bit Animations



Game Selector

TRUE FORCE GAMES



POWER PULLER

CLASSIC MODE

1ST

000

2ND

000

3RD

000

CURRENT FORCE

000

LB

PULL TO START

SESSION BEST

000

TOP SCORE

000

LAST HIT

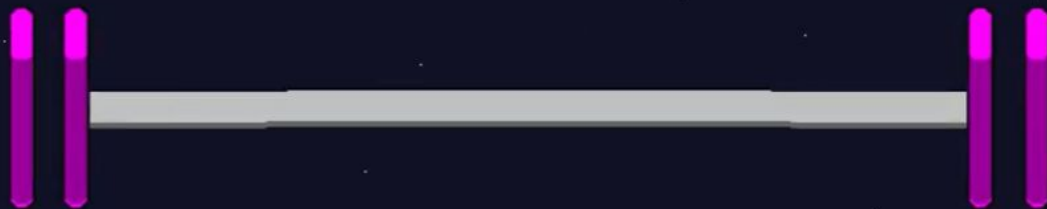
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SPACE START W/UP PULL S/DOWN DROP

LOCAL PREVIEW

BACK

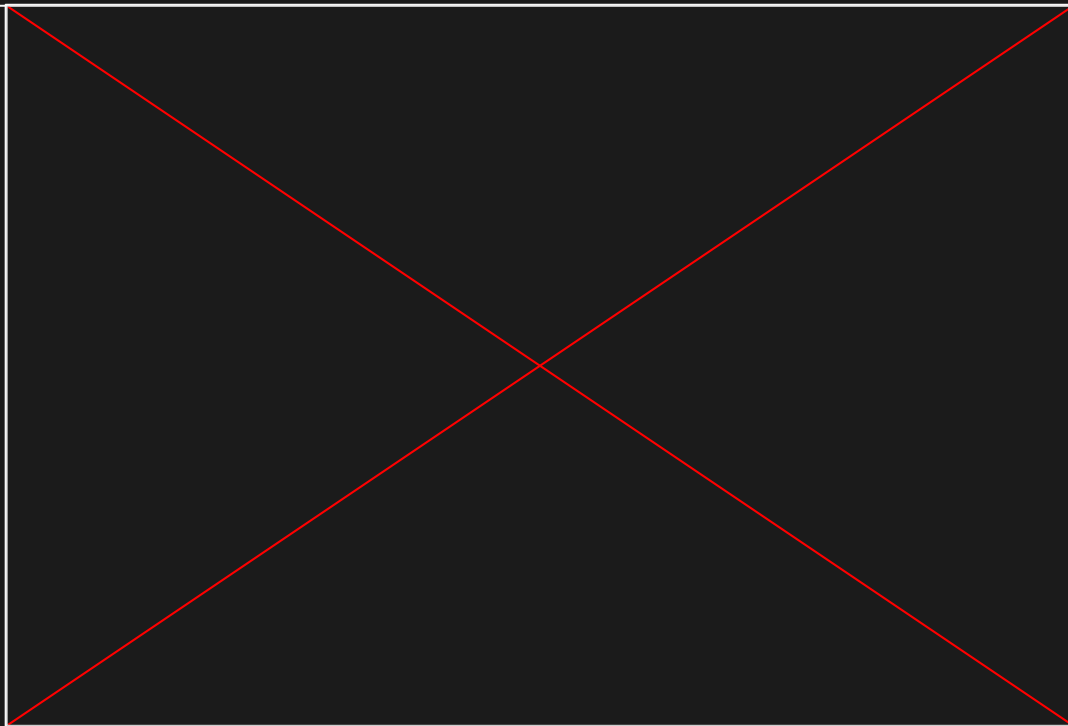
POWER PULLER 3000



CURRENT: 35 lb BEST: 0 lb

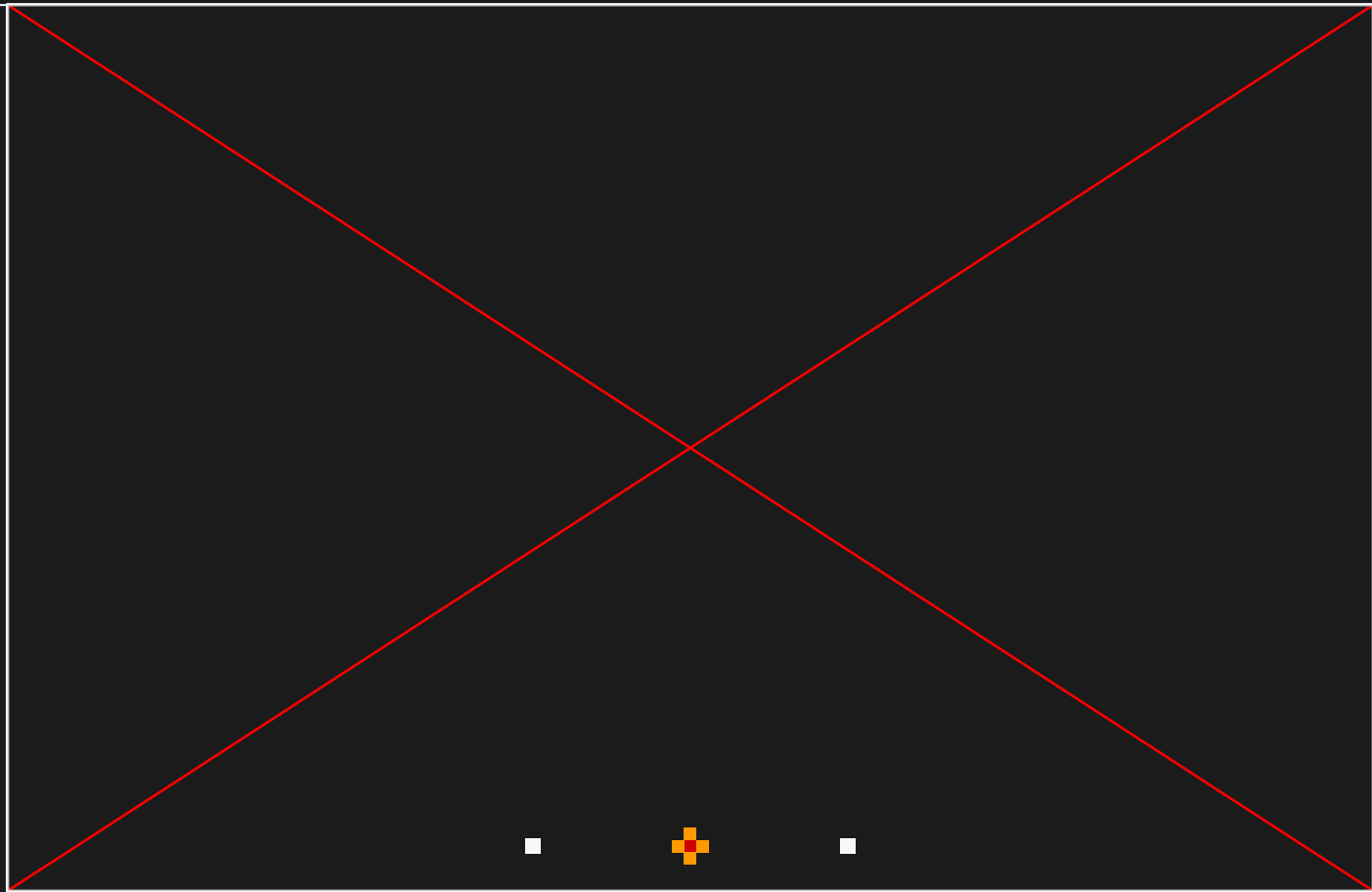
BACK

Power Puller 3000





Rocket Launch



Rocket Launch



Brains VS Brawns

BRAINS VS BRAWN

3 Rounds - Lowest Score Wins

Push Play to Begin

BACK

Brains VS Brawns

BRAINS VS BRAWN

3 Rounds - Lowest Score Wins

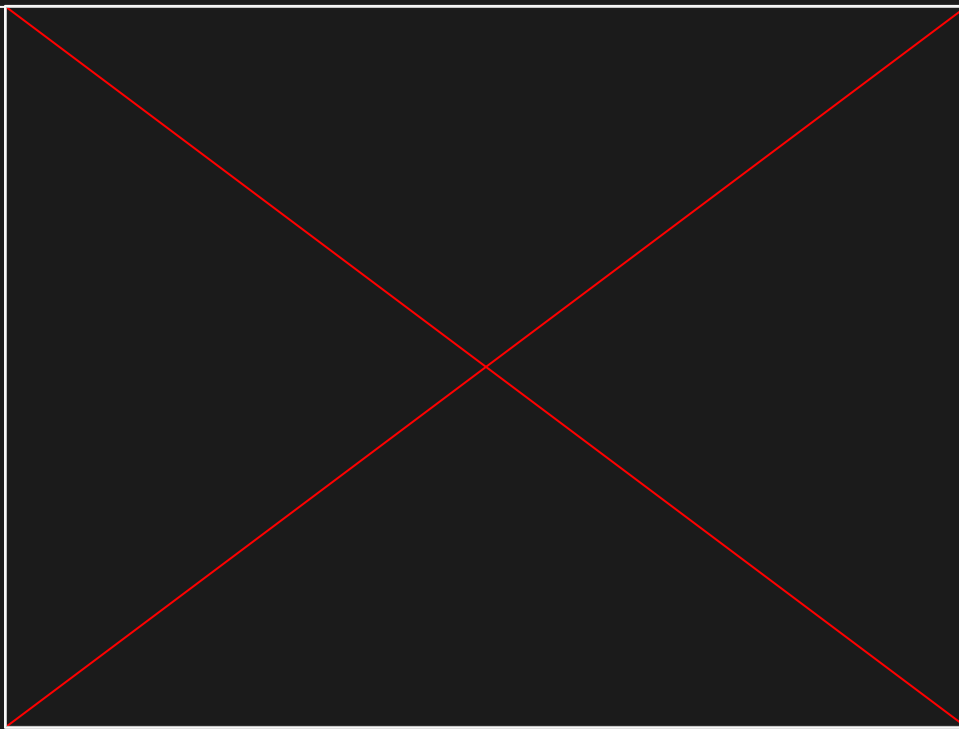
Push Play to Begin

BACK

UFO Escape



UFO Escape



Power Ups:



Obstacles:



Score:

- Distance x cows caught



Balloon Burst



Balloon Burst



Current State

- Minigames are functional
- Hardware is functional
- Force measurement integrated
- User tested
- User experience is complete



Next Steps

- Exterior Visuals
- Receipt Printer
- LEDs
- Audio Implementation
- Raising the bar



Lessons Learned

- Communication with client and teammates
- Flexibility with project direction
- Stepping out of our comfort zone



Project Graveyard (Things Dylan Broke)

2x
Raspberry
Pi

3x HX711
Amplifiers

1 Load
Sensor

1 Pi Hat



Thank You For Listening!

Questions?

